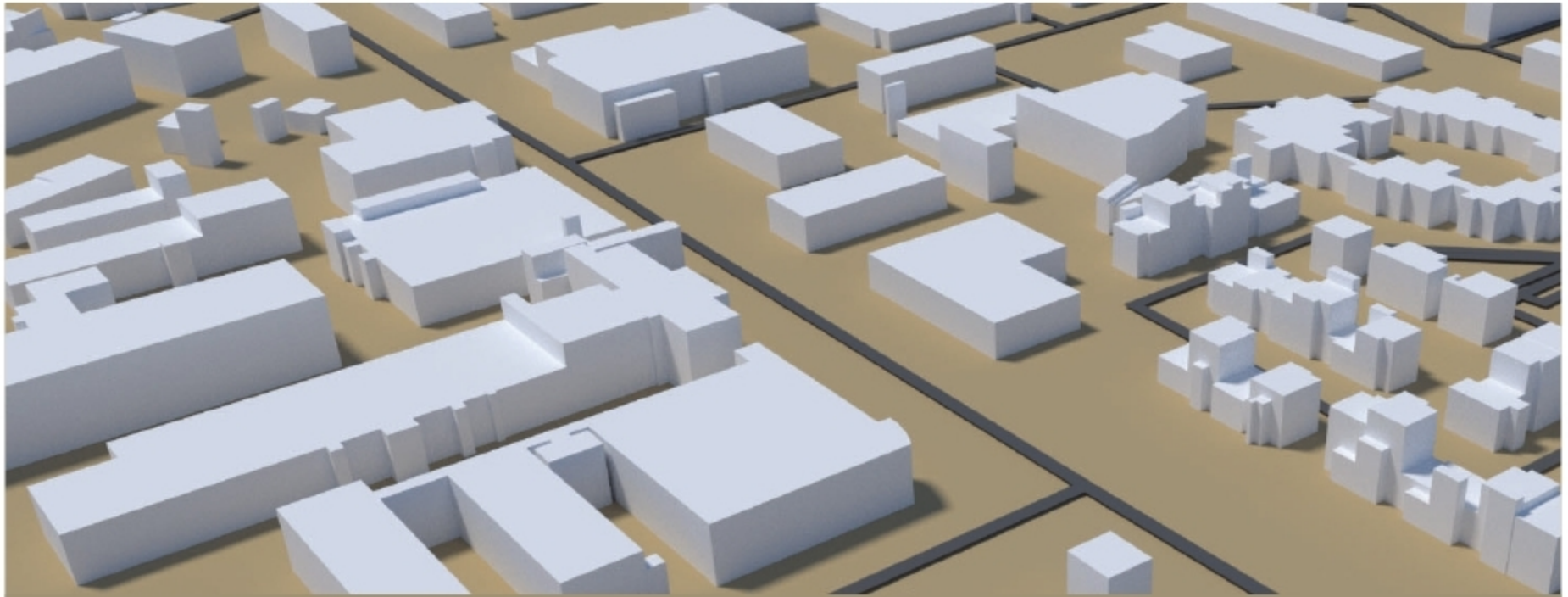


Simulating Smart Cities with CupCarbon



CupCarbon is a smart city and IoT wireless sensor network (WSN) simulator. It is a new platform for 2D/3D design, visualisation and the simulation of radio propagation and interferences in IoT networks. It is particularly relevant in India today, since the development of smart cities is a priority of the government.

It was a wide range of devices interconnected through wireless technologies that gave birth to the Internet of Things (IoT). A number of smart gadgets and machines are now monitored and controlled using IoT protocols. Across the world, devices enjoy all-time connectivity because of the IoT.

From the research reports of *Statista.com*, sales of smart home devices in the US went up from US\$ 1.3 billion to US\$ 4.5 billion, from

2016 to 2019. The *Economic Times* reports that there will be around 2 billion units of eSIM based devices by 2025. An eSIM enables subscribers to use the digital SIM card for smart devices and the services can be activated without a physical SIM card. It is one of the recent and more secure applications of IoT.

Beyond the traditional applications, IoT is being researched for purposes like monitoring the environment

and providing prior notifications to regulating agencies so that appropriate action can be taken, when required. Reports from *LiveMint.com* indicate that the Indian Institute of Technology, New Delhi and Ericsson are partnering to tackle the air pollution in Delhi. News reports from Grand View Research Inc. indicate that the global NB (Narrow Band)-IoT market size is predicted to touch more than US\$ 6 billion by 2025. NB-IoT refers to the radio technology



Figure 1: Key elements and components of a smart city project



Figure 2: Official Web portal of the CupCarbon simulator